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a) *Cycas*


b) *Thuja*


c) *Taxus*


d) *Ginkgo*

Figure 2.38: Gymnosperms

2.6.2 Classification of Gymnosperms

Sporne (1965) classified gymnosperms into 3 classes, 9 orders and 31 families. The classes include i) Cycadospsida ii) Coniferopsida iii) Gnetopsida.

GYMNOSPERMS		
Class-I	Class-II	Class-III
Cycadopsida	**Coniferopsida**	**Gnetopsida**
Orders: 1. Pteridosper-males 2. Bennettitales 3. Pentoxylales 4. Cycadales	Orders: 1. Cordaitales 2. Coniferales 3. Taxales 4. Ginkgoales	Order: 1. Gnetales

General Characters of Main classes:

Class I – Cycadopsida

- Plants are palm-like or fern-like.
- Compound, frond-like pinnate leaves.
- Manoxylic wood.
- Sperms are motile.
- Flower like structures are absent. Strobili are simple.

  Example: *Cycas, Zamia*.

Class II – Coniferopsida

- Tall trees with simple leaves of varied shape.
- Wood is pycnoxylic.
- Cone like strobili are present.
- Motile sperms are absent (except *Ginkgo biloba*). Example: *Pinus*.

Class III – Gnetopsida

- Shrubs, trees and lianas.
- Leaves are elliptical or strap-shaped, simple, opposite or whorled.
- Motile sperms are absent.
- Wood contains vessels.
- Strobili is called as inflorescence.
- Flower like structure with perianth is present. Example: *Gnetum, Ephedra*.

2.6.2 Comparison of Gymnosperm with Angiosperms

Gymnosperms resemble with angiosperms in the following features

- Presence of well organised plant body which is differentiated into roots, stem and leaves.
- Presence of cambium in gymnosperms as in dicotyledons.
- Flowers in *Gnetum* resemble to the angiosperm male flower. The Zygote represent the first cell of sporophyte.
- Presence of integument around the ovule.
- Both plant groups produce seeds.
- Pollen tube helps in the transfer of male nucleus in both.
- Presence of Eustele.